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PI Name:	Goree, John Ph.D.		
Project Title:	Self-Structuring in Dusty Plasmas (NNX07AD22G)		
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Comments:			
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Flight Program:			
Flight Assignment:	NOTE: Received funding continuation to 4/14/2013, per NSSC (8/2010) NOTE: Received extension to 2/28/11 (4/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Morfill, Gregor (Max Planck Institute for Extraterrestrial Physics) Thomas, Hubertus (Max Planck Institute for Extraterrestrial Physics)		
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Task Description:

This fundamental research is an experimental study of dusty plasma physics.
Dusty plasma consists of particles of solid matter suspended in ionized gas. Micron-size particles acquire a large negative charge of typically 10,000 elementary charges. Due to inter-particle repulsion, they arrange themselves in structure with solid or liquid-like properties, similar to a colloidal crystal. With particles of this size, gravity is a dominant force, causing particles to sediment to a thin layer at the bottom of the plasma.

Under microgravity conditions, particles fill a 3D volume and the suspension self-organizes. We study:

• Structural features. These include the boundaries of the particle suspension, including a void at the plasma's center and vortices at its edges.

• Dynamical behavior exhibited by the dusty plasma. These include waves, instabilities, and transport. This dynamical behavior is studied at an atomistic level, using video microscopy to observe the motion of microspheres which serve as an analog for atoms in normal molecular matter.

Experimental methods include direct imaging of particle suspensions using video microscopy and particle tracking, and manipulation of particles using the radiation-pressure force of powerful cw lasers.

Work on this project include experiments performed on the ISS (International Space station), laboratory experiments to prepare for future space missions, and numerical simulations.

Two European instruments on the International Space Station (ISS) are used. First, the PK-3 Plus instrument is currently orbit (the acronym PK refers to "Plasma Crystal"). Second, the PK-4 instrument is in its final engineering stages. Ground-based laboratory experiments are carried out to prepare for future microgravity flight experiments, including the planned PK-4 instrument. NASA is not expected to provide any flight hardware. Video cameras are used to image not only the particle suspension as in previous experiments, but also spectral emission from the plasma glow. These space-flight experiments on ISS are carried out under the direction of German and Russian scientists. The PI is a team member on both the PK-3 Plus and PK-4 missions, and receives the data after it is returned from the ISS.

Ground-based laboratory experiments and numerical simulations accompany the analysis of data returned from space. This ground-based work serves the dual purpose of developing scientific methods to be used in future flight experiments as well as providing an immediate scientific return.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Status of ISS missions:
The PI, Professor Goree, is a team member for two missions on the International Space Station (ISS): The PK-3 Plus instrument is now on orbit and returning data. This instrument is primarily a German-Russian collaboration. The PK-4 mission is primarily a European Space Agency (ESA) mission, and is now in the Phase D engineering stage. The acronym PK refers to the German spelling of "Plasma Crystal." Both projects concern dusty plasma, also known as complex plasma, which consists of micron-size particles of solid matter immersed in ionized gas.

Analysis of ISS data: During this period, Professor Goree received one shipment of new data from PK-3 Plus, in addition to previous shipments already received. Data were returned aboard Soyuz spacecraft, and later delivered on computer hard disks to Professor Goree in Iowa. The data consist of digitally-recorded video accompanied by time series recordings of voltage, temperature, and other housekeeping data. After receiving these shipments, the PI now has five hard disks of data from PK-3 Plus.

Analysis of PK-3 Plus microgravity experiment data has been carried out, and remains underway. This work is being done by two members of Professor Goree's group. Two main scientific topics were explored at Iowa using PK-3 Plus data during this period: (1) using particle imaging as a diagnostic for charge, electron temperature and ion density, and (2) a study of the basic physics topic of non-Gaussian statistics for random particle motion.

In the first topic, we use a combination of static measurements of particle positions and dynamic measurements of transverse waves in the suspensions of microparticles in order to measure three quantities that cannot be directly measured by other diagnostics on board PK-3 Plus. The dust particle charge is perhaps the most essential parameter, yet experimenters for PK-3 Plus had no way to measure it in-situ, with the plasma on. To face this challenge, we devised a method which we have now reported in Physics of Plasmas, to measure the particle charge based on the observed resonant frequency for out-of-plane particle motion in a single-layer suspension at the edge of a void. We then extended this method to yield measurements of two other parameters which cannot be measured directly in PK-3 Plus: the temperature of electrons and the number density of ions, in the ambient plasma that fills the dusty plasma chamber. This work was published in 2010 in Physics of Plasmas. Russian and German team members were co-authors.

In the second topic, which is underway, we investigate non-Gaussian statistics, which is a fundamental statistical physics problem that can take great advantage of the unique data provided by dusty plasma experiments. Using data from a microgravity experiment with no macroscopic flows, we analyze the microscopic random motion of the charged dust particles, as they interact with one another. We calculate the velocity distribution function, and we find that it has "fat tails," i.e., an overabundance of energetic particles as compared to a Gaussian or Maxwell-Boltzmann distribution. We fit the measured distribution to the Tsallis distribution, which is a theoretical expression intended for non-equilibrium systems, and we find a good fit. Such a result was found previously in laboratory experiments under conditions with high-speed ion flows; here, microgravity conditions enabled experiments with low-speed ions. (This is because in the presence of gravity, a large dc electric field is required to levitate the dust particles, and this large electric field drives a fast ion flow, while under microgravity conditions the electric field can be orders of magnitude weaker.) Because ion flow past the charged dust particles is a non-equilibrium condition, signatures of non-Gaussian statistics are expected, but these signatures are expected to be weaker with low-speed ions (in microgravity conditions) than with high-speed ions (in laboratory conditions). Surprisingly, we find statistically significant deviations from Gaussian statistics, even with the low-speed ion flow under microgravity conditions.

Ground-based experiments:

Task Progress:	During this period, Professor Goree's group also carried out ground-based laboratory research to support the flight experiments. Microgravity enables the study of three-dimensional suspensions of microparticles, unlike laboratory conditions where the particles usually sediment to form a thin horizontal layer. The first three laboratory experiments were studies of the dust density wave, which is a low-speed compression wave that is observed in PK-3 Plus as well, so our laboratory experiments help prepare for future analysis of PK-3 Plus data by identifying experimental methods and data analysis methods. Three dust-density wave experiments were completed and reported: • The first experiment was a study of the onset of an instability that drives the waves, and it was carried out using high-speed video imaging and a very careful slow variation of gas pressure. Gas pressure is significant because gas provides a damping mechanism that offsets the energy source of the waves, due to ion flow. By making very small changes in gas pressure, we were able to observe the onset of the instability, and characterize the wave's frequency, wave number and spatial growth rate. We also developed a theoretical dispersion relation to compare to the experimental observations. These results were published in 2010 in Physics of Plasmas. • The second experiment was a study of nonlinearities in the dust-density wave. In this second project, we introduced several innovations to the study of nonlinear plasma waves. We focused our attention on harmonics of the fundamental wave as the signature of nonlinearity. Using our recently-developed phase-sensitive detection method, we found that the wavenumber and growth rate of each harmonic was surprisingly a simple integer multiple of those for the fundamental. We also introduced the total-harmonic distortion (THD) as a measure of wave nonlinearity, exploiting our measurements that were made possible by the phase-sensitive detection. Results from this second experiment have been submitted for publication in Physics of Plasmas. • The third dust-density wave experiment was carried out in collaboration with two leading German researchers, Prof. Alexander Piel and Dr. Oliver Arp, of the University of Kiel. The results, published in IEEE Transactions on Plasma Science, reveal that the wave properties are more complicated than previously known because they involve a modulation of the energy of the ambient electrons, in addition to modulating the number density of dust particles. Additional laboratory experiments were carried out to study melting, viscoelasticity, and wave coupling. These phenomena can also be studied under microgravity conditions for a 3D suspension. Here, we used a single-layer suspension to study the 2D physics that corresponds to the 3D problems. • Melting was studied using a new analysis method, polygon analysis, which was developed originally by colloid physics investigators. This project remains underway, and is expected to lead to new methods of analyzing data from the microgravity experiments. • Viscoelasticity was studied by observing the random motion of particles and computing the wavenumber-dependent viscosity; these results were published in Physical Review Letters. • The wave coupling phenomenon involves two different kinds of waves, compressional and transverse, that become mixed into a single wave due to the streaming of ions past the charged dusty particles. These results were published in Physical Review Letters. We are now examining data from PK-3 Plus to search for similar phenomena, under microgravity conditions. Numerical simulations: Numerical simulations were carried out to study viscoelasticity. These simulations were carried out in collaboration with two leading Hungarian scientists, Dr. Zoltan Donko and Dr. Peter Hartmann. The results, published in Physical Review E, are suitable for direct comparison to experiments. Planning of future space missions: During 2010, the PI participated in ESA's meeting of the PK-4 Facility Science Team in St. Petersburg, Russia. Serving as Vice-Chairman of the team, the PI ran the meeting. During this meeting, input from scientists was collected to guide ESA during the engineering Phase C/D for this mission.

Bibliography Type:	Description: (Last Updated: 01/23/2025)
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